



Hydraulic System

Hydraulic Valve

Model: FX-2652C

OPERATION AND MAINTENANCE MANUAL

Version: 1.0

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TABLE OF CONTENTS

Section	Page
Table Of Contents	2
Introduction	5
Safety Instructions	6
Technical Specifications	7
Installation	10
Operation	13
Maintenance	16
Troubleshooting	19
Parts List	20
Wiring Diagrams	23
Standards	25
Appendix	27

All personnel involved in installation, operation, or maintenance of this equipment must be properly trained and familiar with the contents of this manual.

This manual is organized into sections covering safety, technical specifications, installation, operation, maintenance, troubleshooting, and parts information. Each section should be reviewed before beginning work.

Safety is the highest priority when working with industrial equipment. Always follow safety instructions and use appropriate personal protective equipment.

For technical support or additional information, please contact your local distributor or visit our website.

The information in this manual is based on the latest technical data available at the time of publication. Specifications and procedures may be updated periodically to reflect improvements and changes.

This manual provides comprehensive information for the operation, maintenance, and troubleshooting of the Hydraulic System model FX-2652C.

The Hydraulic System is designed for industrial applications requiring reliable performance and long service life. Proper installation, operation, and maintenance are essential for achieving optimal results.

Warning	Description	Elimination/Action
	Allow equipment to cool before maintenance. Use appropriate protective equipment when handling hot components.	Wait for equipment to reach ambient temperature. Wear heat-resistant gloves.
	Some lubricants and materials may cause allergic reactions in sensitive individuals.	Use protective gloves. Ensure adequate ventilation. Consult material safety data sheets.
	Pressure can build up during operation. Sudden release can cause injury.	Release pressure slowly using appropriate procedures. Never remove caps or plugs under pressure.
	Overfilling can cause leakage, overheating, and equipment damage.	Follow manufacturer specifications for fill levels. Check levels regularly.
	Mixing incompatible oils can cause equipment failure and void warranty.	Use only specified lubricants. Drain completely before changing oil type.
	Warming oil before use improves flow and ensures proper distribution.	Heat oil to recommended temperature (typically 40-50°C) before application.
	The amount of new lubricant should match the volume drained to maintain proper levels.	Measure drained volume accurately. Add equivalent amount of new lubricant.
	Residual oil can be slippery and cause falls. Dispose of properly.	Clean up spills immediately. Use appropriate absorbent materials. Follow local disposal regulations.

All specifications are subject to change without notice as improvements are made to the product. Always refer to the latest version of this manual for current specifications.

Some specifications may be optional or configurable. Consult with technical support or your local distributor to determine available options for your specific requirements.

Environmental conditions such as temperature, altitude, and humidity can affect equipment performance.
Specifications are based on standard conditions unless otherwise specified.

This section provides detailed technical specifications for the Hydraulic System model FX-2652C. All specifications are based on standard operating conditions unless otherwise noted.

General Specifications

Parameter	Value	Unit	Notes
Pressure Range	420	bar	Optional
Flow Rate	25	m³/h	
Power Range	15	kW	
Operating Temperature	0°C		As specified
Weight	85	kg	Approximate
Operating Temperature	-19 to 60	°C	Continuous
Storage Temperature	-23 to 61	°C	
Humidity	12 to 88	% RH	Non-condensing
Protection Class	IP65		IEC 60529
Noise Level	59	dB(A)	At 1m distance
Vibration	≤ 3	mm/s	RMS
Service Factor	1.15		

Dimensions

Dimension	Value	Unit	Tolerance
Length	877	mm	±2 mm
Width	532	mm	±2 mm
Height	275	mm	±2 mm
Mounting Holes	M8		
Shaft Diameter	42	mm	±0.1 mm
Base Height	198	mm	±1 mm

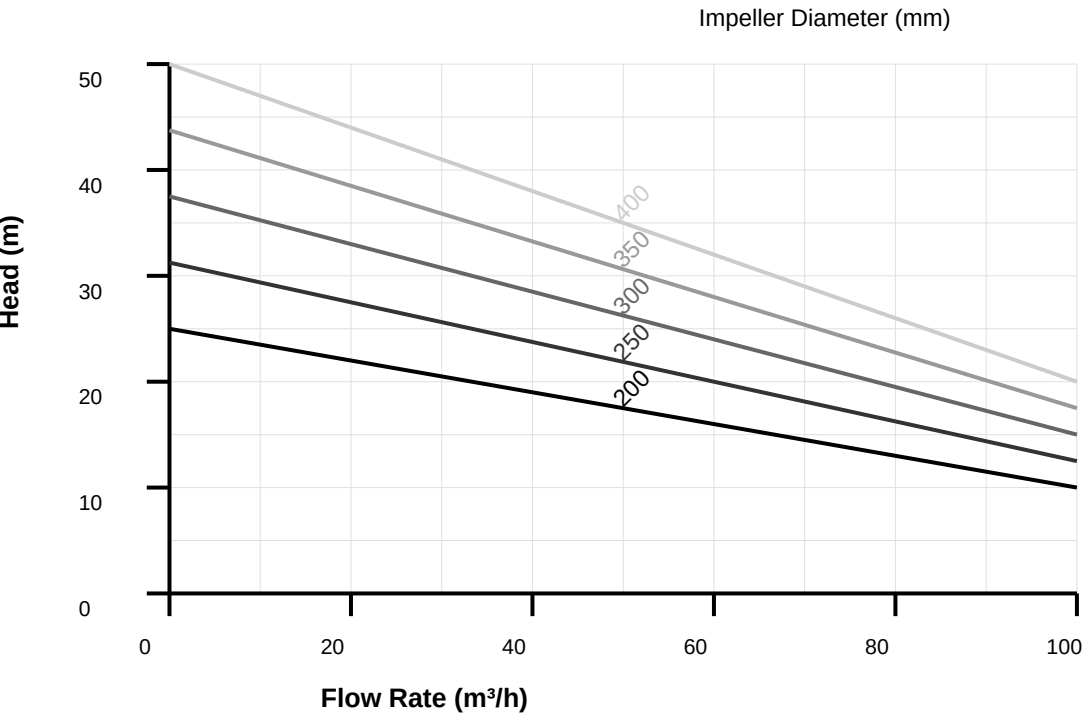
Performance Data

Operating Condition	Minimum	Nominal	Maximum	Unit
Power Consumption	80	95	110	%
Efficiency	88	95	99	%
Load Factor	23	77	102	%

Operating Condition	Minimum	Nominal	Maximum	Unit
Speed	720	1483	2960	rpm
Torque	33	79	160	Nm

Environmental Conditions

Condition	Operating Range	Storage Range	Unit
Temperature	-18 to 40	-26 to 69	°C
Relative Humidity	16 to 85	7 to 91	%
Altitude	0 to 1397	0 to 2585	m
Vibration	≤ 2	≤ 10	mm/s



Electrical Specifications

Parameter	Value	Unit	Standard
Supply Voltage	400V	V AC	IEC 60038
Frequency	50 Hz	Hz	IEC 60034
Current Rating	80	A	
Power Factor	0.95		

Parameter	Value	Unit	Standard
Starting Current	433	% of rated	
Insulation Class	H		IEC 60085
Protection	IP54		IEC 60529

Regular inspection of electrical connections and components is essential for safe and reliable operation.
Include electrical inspection in your maintenance schedule.

Protection class ratings indicate the degree of protection against ingress of solid objects and water. Ensure that the protection class is appropriate for your installation environment.

4.1 Installation Requirements

Installation Procedures

Action	Note
- Lubricate the sealing with grease just before fitting. (Not too early - there is a risk of dirt and foreign particles adhering to the sealing.) - Fill 1/2 of the space between the dust lip and the main lip with grease. - If the sealing is without dust lip, just lubricate the main lip with a thin layer of grease.	Ensure that no grease is applied to the marked surface. Use only approved lubricants specified in the parts list.
- Check the sealing to ensure that: - The sealing is of the correct type. - There is no damage on the main lip.	Article number is specified in Equipment on page 156.
- Inspect the shaft surface before mounting. - If scratches or damage are found, the shaft must be replaced since it may result in future leakage. - Do not try to grind or polish the shaft surface to get rid of the defect.	Ensure proper surface finish is maintained. Refer to technical specifications for surface roughness requirements.
- Apply proper torque to all fasteners. - Use torque values specified in the technical data sheet (23 Nm for main bolts). - Tighten in the correct sequence as shown in the installation diagram.	Overtightening can damage components. Undertightening can cause loosening during operation.

Some installation tasks may require specialized tools or equipment. Ensure all necessary tools are available before beginning.

Auxiliary systems such as cooling, lubrication, and ventilation must be properly connected and verified before startup.

After installation, perform all required alignment, calibration, and verification procedures before placing the equipment into service.

Coordinate with other trades and systems to ensure proper integration of the equipment into the overall facility.

The Hydraulic System requires careful handling during transport and installation to prevent damage to critical components.

Before installation, ensure the following requirements are met:

Installation Checklist

Requirement	Specification	Status
Adequate space for installation and maintenance access	As per installation manual	■
Proper foundation or mounting surface	As per installation manual	■
Correct electrical supply voltage and frequency	As per installation manual	■
Appropriate environmental conditions (temperature, humidity)	As per installation manual	■
Compliance with local building and electrical codes	As per installation manual	■

Foundation Requirements

Parameter	Minimum	Recommended	Unit
Foundation Thickness	312	351	mm
Concrete Strength	C21	C29	
Reinforcement	As per local code	As per local code	
Anchor Bolt Size	M12	M16	
Anchor Bolt Depth	190	273	mm

4.2 Installation Steps

Installation Procedure

Step	Description	Duration	Tools Required
1	Unpack the equipment and inspect for damage.	15 min	Multimeter
2	Position the equipment according to layout drawings.	15 min	Alignment tools, Multimeter
3	Secure the equipment to the foundation using appropriate fasteners.	30 min	Multimeter, Wrench set, Torque wrench
4	Connect electrical wiring according to wiring diagrams.	30 min	Level

Step	Description	Duration	Tools Required
5	Connect auxiliary systems (cooling, lubrication, etc.).	30 min	Alignment tools
6	Perform initial alignment and calibration.	30 min	Alignment tools
7	Verify all connections and safety systems.	30 min	Alignment tools, Level, Torque wrench
8	Perform initial test run according to startup procedure.	1 hour	Level, Multimeter

Torque Specifications

Component	Bolt Size	Torque	Unit
Mounting Bolts	M12	83	Nm
Shaft Coupling	M8	38	Nm
Terminal Connections	M6	8	Nm
Cover Bolts	M8	18	Nm

During installation, periodically verify that all connections are secure and that no damage has occurred to components. Address any issues before proceeding.

If installation issues arise, stop work and consult with technical support before proceeding. Do not force components or make unauthorized modifications.

Follow the installation steps in the order specified. Skipping steps or performing them out of sequence may result in improper installation or equipment damage.



Note

After installation, allow sufficient time for the equipment to stabilize before performing calibration or alignment procedures.

For more information see:

- **Installation Guide - Electrical Connections**

5.1 Operating Procedures

Some operating modes may have specific requirements or limitations. Review operating mode specifications before changing modes.

Monitor operating parameters regularly to ensure the equipment is operating within specified limits. Abnormal readings may indicate problems requiring attention.

Operators must be trained and familiar with the equipment operation, control functions, and emergency procedures. Never operate equipment without proper training.

If operating conditions change significantly, review operating parameters and adjust as necessary. Sudden changes may indicate problems requiring attention.

Follow the startup sequence exactly as specified. Improper startup procedures may cause equipment damage or safety hazards.

Follow these procedures for safe and efficient operation:

Operating Parameters

Parameter	Setting	Range	Unit
Startup Sequence	Automatic	Manual/Automatic	
Operating Mode	Continuous	Continuous/Intermittent	
Control Method	Manual	Manual/Automatic/Remote	
Monitoring	Enabled	Enabled/Disabled	
Speed Setpoint	1449	2194	rpm
Load Limit	93	38	%

Startup Sequence

Step	Action	Duration	Status Indicator
1	Power On	Immediate	Power LED
2	System Initialization	15	Init LED
3	Self-Test	20	Test LED
4	Ready State	Continuous	Ready LED
5	Start Command	On Demand	Run LED

5.2 Control Functions

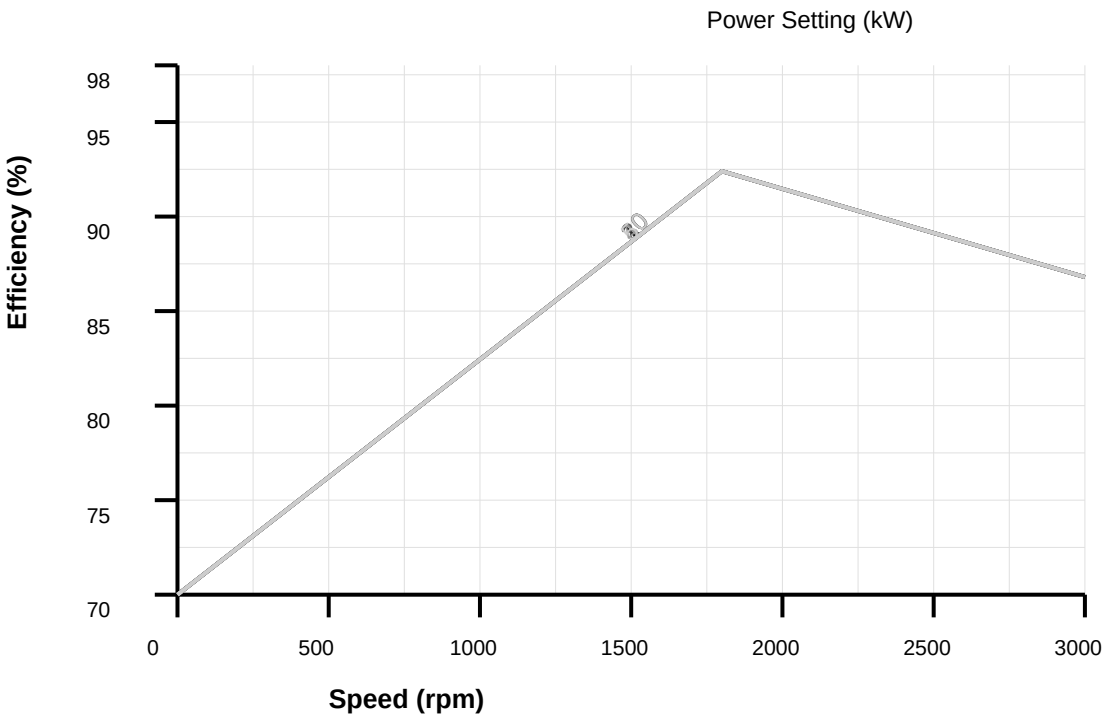
Control Functions

Control	Function	Location	Type
Start/Stop	Controls equipment operation	Panel A	Push Button
Speed Control	Adjusts operating speed within specified range	Panel B	Selector Switch
Emergency Stop	Immediately stops all operations	Panel C	Emergency Stop
Reset	Clears fault conditions and resets system	Panel D	Reset Button
Mode Selection	Selects operating mode	Panel E	Selector

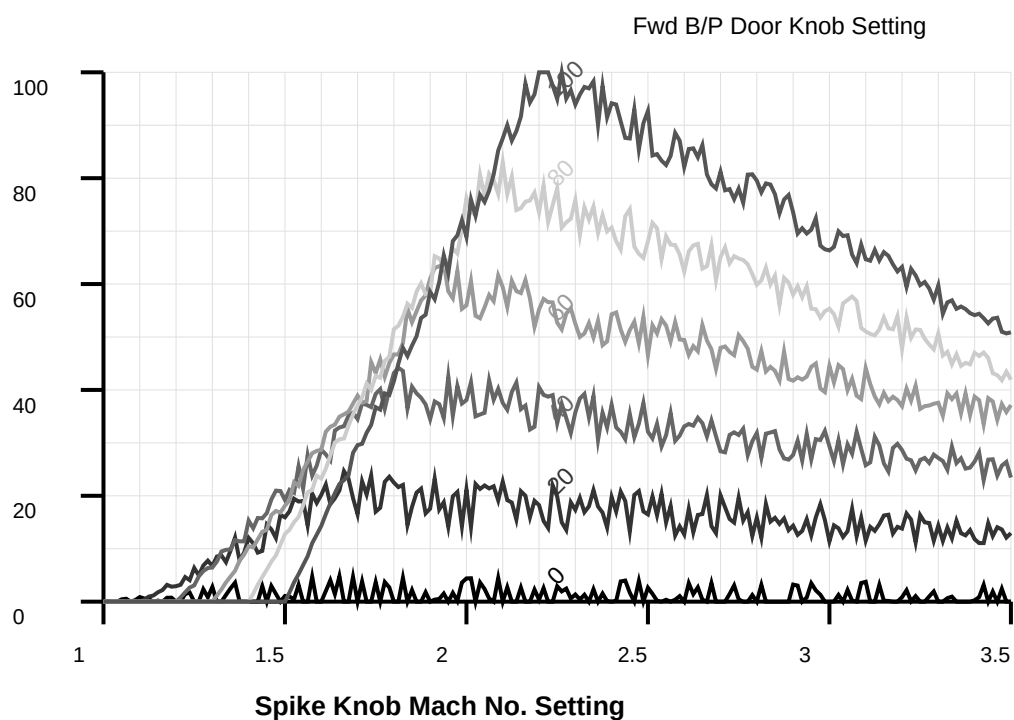
Status Indicators

Indicator	Color	Meaning	Action
Power	Green	System powered	Normal operation
Run	Green	Equipment running	Normal operation
Fault	Red	Fault detected	Check fault code
Warning	Yellow	Warning condition	Review warning
Maintenance	Yellow	Maintenance due	Schedule service

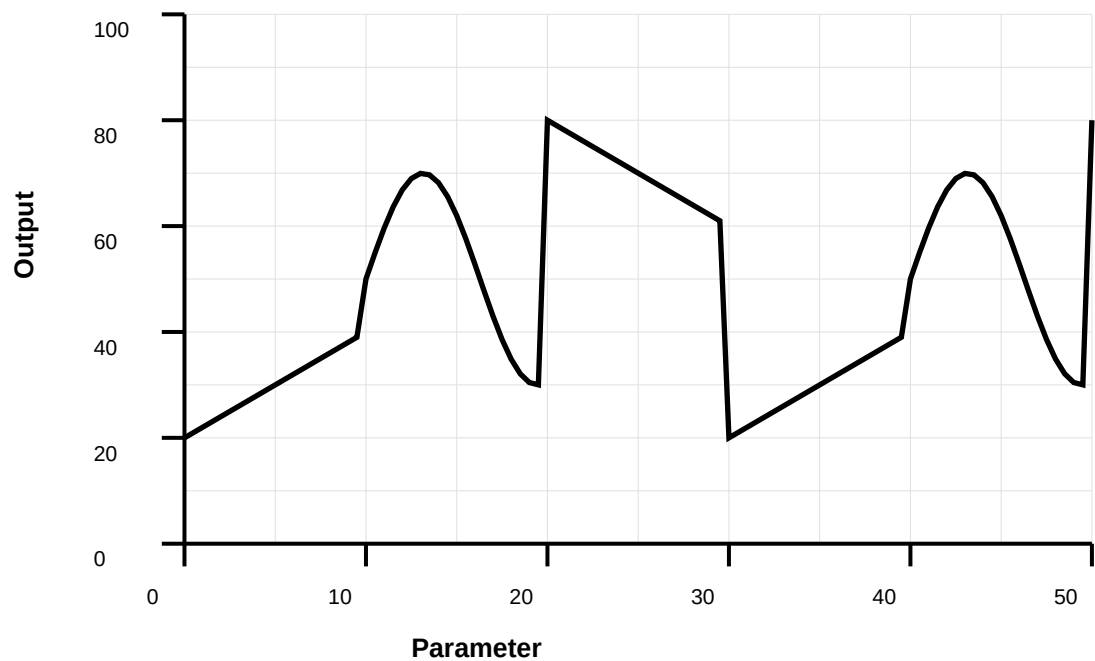
Operating Characteristics



Indicated Fwd B/P Door Opening ~ % Open



Bifurcation Diagram



Emergency stop functions must be tested regularly to ensure they are functioning properly. Never disable safety systems.

When operating in automatic mode, verify that all control signals are functioning correctly and that the equipment responds appropriately to commands.

Keep operating logs documenting operating hours, parameters, and any issues encountered. This information is essential for maintenance planning and troubleshooting.

6.1 Maintenance Schedule

Before performing any maintenance, ensure the equipment is safely shut down and all safety protocols are followed. Refer to Section 2 for safety instructions.

Keep detailed maintenance records including dates, tasks performed, parts replaced, and lubricants used. This information is required for warranty claims and service history.

All maintenance work must be performed by qualified personnel familiar with the equipment and maintenance procedures. Improper maintenance may cause equipment damage or safety hazards.

Use only approved replacement parts and lubricants specified in this manual. Using incorrect parts or lubricants may cause equipment failure and void warranty.

Maintenance Schedule

Maintenance Task	Frequency	Duration	Personnel
Visual Inspection	Daily	5 min	Operator
Lubrication	Every 208 hours	30 min	Technician
Filter Replacement	Every 840 hours	1 hour	Technician
Bearing Inspection	Every 1239 hours	2 hours	Engineer
Complete Overhaul	Every 7587 hours	8 hours	Service Team
Calibration Check	Every 2724 hours	1 hour	Engineer

Inspection Checklist

Item	Check	Acceptance Criteria	Frequency
Bearings	Visual/Temperature	No noise, temp < 70°C	Weekly
Seals	Visual	No leakage	Weekly
Mounting	Visual/Torque	Bolts tight, no movement	Monthly
Electrical	Visual/Measurement	No damage, correct voltage	Monthly
Cooling	Visual/Temperature	Clean, adequate flow	Weekly
Vibration	Measurement	< 5 mm/s RMS	Monthly

6.2 Maintenance Procedures

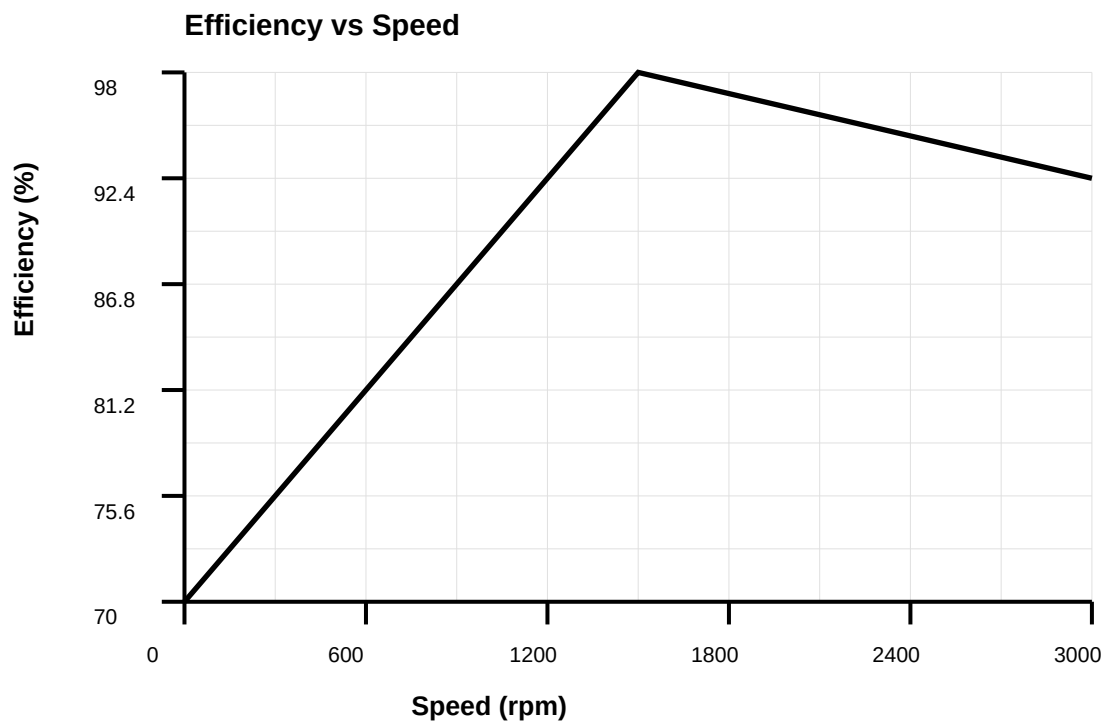
- Always shut down and lock out equipment before maintenance.
- Follow manufacturer's recommendations for lubricants and spare parts.
- Inspect all components for wear, damage, or contamination.
- Replace worn or damaged parts immediately.
- Keep maintenance records for warranty and service history.
- Use only approved replacement parts and materials.

Component Maintenance Intervals

Component	Inspection Interval	Replacement Interval	Condition
Bearings	Every 1287 hours	Every 11964 hours	Based on condition
Seals	Every 733 hours	Every 3176 hours	If leaking
Filters	Every 821 hours	Every 1199 hours	If dirty
Belts	Every 1415 hours	Every 3272 hours	If worn

After completing maintenance, verify that all components are properly installed and that the equipment is ready for operation before restarting.

Some maintenance tasks require specialized tools or training. Do not attempt maintenance beyond your qualification level. Contact qualified service personnel when needed.



**Note**

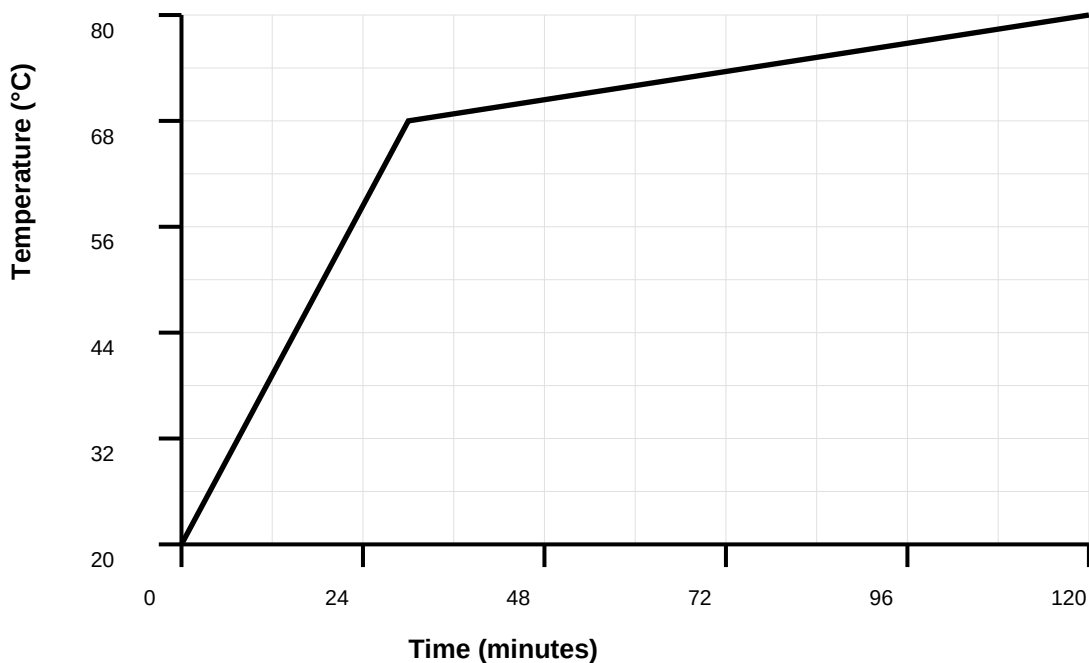
Inspect components regularly for signs of wear or damage. Early detection of problems can prevent more serious failures and reduce downtime.

For more information see:

- **Maintenance Guide - Lubrication Specifications**

Temperature Characteristics

Temperature Rise vs Time



Keep detailed records of any troubleshooting procedures performed, including symptoms observed, measurements taken, and actions attempted. This information is valuable for warranty claims and future service.

Troubleshooting the Hydraulic System requires systematic diagnosis of symptoms and their underlying causes. This section provides guidance for identifying and resolving common operational issues.

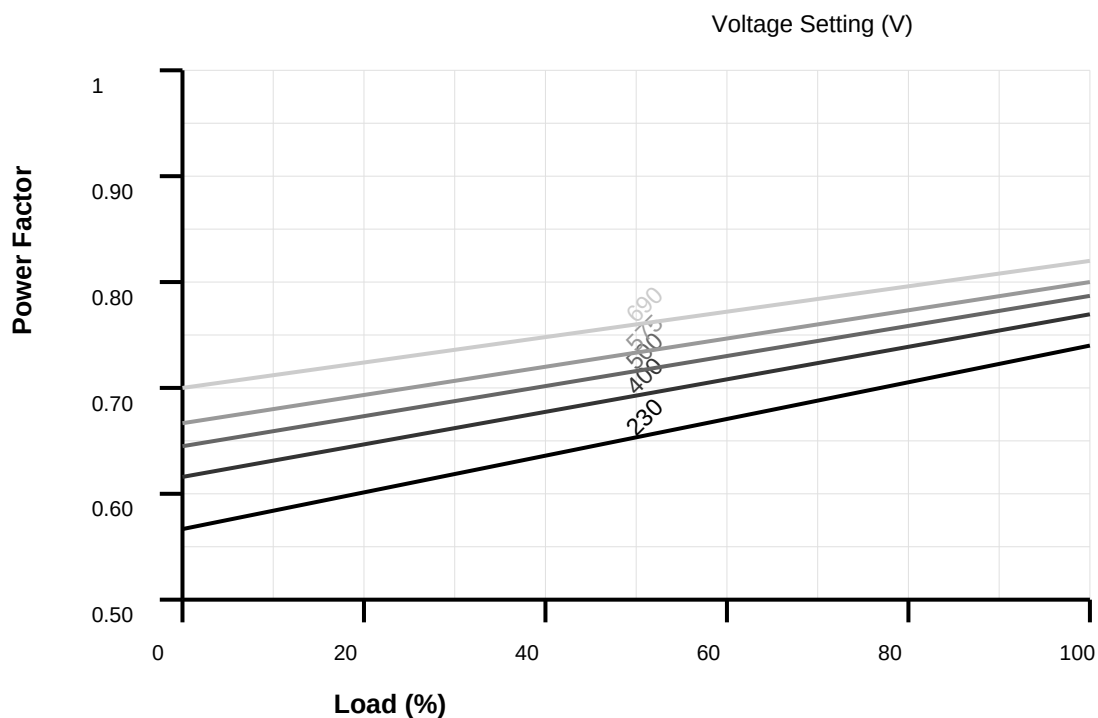
Many issues can be resolved through basic inspection and maintenance. However, some problems may indicate more serious conditions that require professional service. When in doubt, contact technical support.

Common Issues and Solutions

Symptom	Possible Cause	Solution
Ground fault detected	Insulation failure, moisture ingress, or damaged wiring	Measure insulation resistance, inspect wiring for damage, check for moisture
Erratic operation or speed fluctuations	Control system issue or sensor malfunction	Check control system connections, verify sensor readings
Fault Code H426: Low pressure	Pump failure, leak in system, or relief valve issue	Check pump, inspect for leaks, test relief valve
Premature component failure	Incorrect operation, poor maintenance, or environmental factors	Review operating procedures, improve maintenance schedule, check environmental conditions
Equipment will not start	Power supply issue or circuit breaker tripped	Check electrical connections, verify voltage, reset circuit breaker if necessary
Fault Code H971: Pump cavitation	Insufficient inlet pressure, blocked filter, or air ingress	Check inlet pressure, inspect filter, remove air
Seal leakage detected	Seal wear, improper installation, or excessive pressure	Inspect seals for wear, verify installation, check pressure limits
Alarm A70: Fluid level low	Leak in system or consumption	Check for leaks, add fluid if needed
Phase loss or imbalance	Loose connection, damaged cable, or supply issue	Check all phase connections, measure phase voltages, inspect cables
Reduced efficiency over time	Normal wear, contamination buildup, or calibration drift	Perform maintenance, clean components, recalibrate systems
Unexpected shutdown during operation	Protection activation, power loss, or system fault	Check protection settings, verify power supply, review fault logs
Reduced performance or efficiency	Worn components, contamination, or incorrect settings	Inspect and clean components, verify operating parameters

When troubleshooting, always start with the simplest possible causes before investigating more complex issues. Many problems can be resolved through basic inspection and cleaning.

Use systematic diagnostic procedures: observe symptoms, identify possible causes, test hypotheses, and verify solutions. Document each step for future reference.



Note

Before replacing components, verify that the root cause has been identified. Replacing parts without addressing the underlying issue may lead to repeated failures.

For more information see:

- **Technical Support - Contact Information**

Spare Parts List

Part Number	Description	Quantity	Material	Supplier Part No.
FX-2652C-321	Main Housing	3	Stainless Steel	SP-61488
FX-2652C-568	Rotor Assembly	3	Cast Iron	SP-83367
FX-2652C-187	Stator Assembly	2	Stainless Steel	SP-64894
FX-2652C-126	Bearing Set	4	Cast Iron	SP-59196
FX-2652C-654	Shaft	4	Plastic	SP-62308
FX-2652C-738	Seal Kit	2	Plastic	SP-64688
FX-2652C-653	Gasket Set	4	Cast Iron	SP-60268
FX-2652C-941	Mounting Bracket	2	Steel	SP-47560
FX-2652C-432	Cooling Fan	2	Plastic	SP-65220
FX-2652C-620	Terminal Box	3	Plastic	SP-60761

Part Number	Description	Quantity	Material	Supplier Part No.
FX-2652C-551	Nameplate	2	Cast Iron	SP-35100
FX-2652C-182	Fastener Kit	4	Plastic	SP-94781
FX-2652C-943	Thermal Protection	2	Bronze	SP-48341
FX-2652C-961	Vibration Sensor	2	Stainless Steel	SP-57990
FX-2652C-827	Oil Filter	2	Stainless Steel	SP-12662
FX-2652C-585	Cooling Coil	1	Bronze	SP-54530
FX-2652C-129	Impeller	2	Stainless Steel	SP-74991

Recommended Spare Parts

Part Number	Description	Recommended Stock	Lead Time
FX-2652C-330	Bearing Set	1-2	2-4 weeks
FX-2652C-377	Seal Kit	1-2	4-6 weeks
FX-2652C-418	Gasket Set	3-5	1-2 weeks
FX-2652C-645	Filter Element	2-3	1-2 weeks
FX-2652C-528	Thermal Fuse	2-3	2-4 weeks
FX-2652C-441	Vibration Sensor	1-2	2-4 weeks
FX-2652C-666	Cooling Fan	1-2	4-6 weeks

Lubrication Requirements

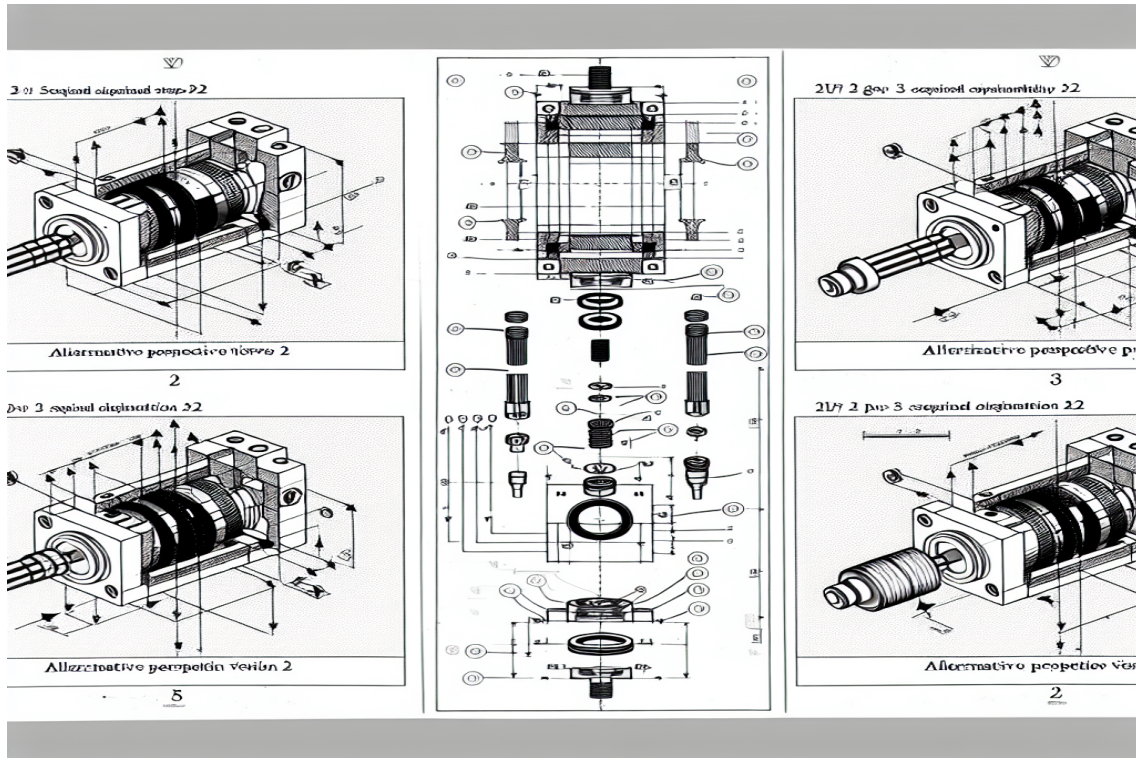
Component	Lubricant Type	Quantity	Interval
Main Bearings	Oil ISO VG 68	154	Every 1014 hours
Gearbox	Oil ISO VG 220	4	Every 3098 hours
Seals	Silicone Grease	22	Every 536 hours

Refer to the wiring diagrams provided in the appendix. All connections must be made according to local electrical codes and regulations.

Terminal Connections

Terminal	Function	Wire Size	Voltage
L1, L2, L3	Power Input	4 mm ²	230V
PE	Protective Earth	2.5 mm ²	0V
U, V, W	Motor Output	6 mm ²	Variable

Terminal	Function	Wire Size	Voltage
24V	Control Supply	0.75 mm ²	24V DC
GND	Control Ground	0.75 mm ²	0V



The Hydraulic System is compliant with applicable international standards for industrial machinery and electrical equipment. In case of deviation from these standards, these are listed in the declaration of incorporation. The declaration of incorporation is part of the delivery.

Machine Standards

Standard	Description
ISO 12100	Safety of machinery - General principles for design - Risk assessment and risk reduction
IEC 60204-1	Safety of machinery - Electrical equipment of machines - Part 1: General requirements

Other Standards Used in Design

Standard	Description
IEC 61000-6-2	Electromagnetic compatibility (EMC) - Part 6-2: Generic standards - Immunity standard for industrial environments
IEC 61000-6-4	Electromagnetic compatibility (EMC) - Part 6-4: Generic standards - Emission standard for industrial environments
ISO 13849-1:2015	Safety of machinery - Safety related parts of control systems - Part 1: General principles for design

A. Technical Data Sheets

Additional technical data sheets and detailed specifications are available upon request.

Revision History

Revision History

Revision	Description
Y	- Published in release R23A. The following updates are made in this revision: - Updated information about special treated screws. - Removed information about position switches as they are no longer available. - Updated torque specifications for various components.
U	- Published in release R24A. The following updates are made in this revision: - Updated information about special treated screws. - Removed information about position switches as they are no longer available. - Updated torque specifications for various components.
Z	- Published in release R22A. The following updates are made in this revision: - New touch up color Safety Yellow available. - Levelmeter 2616 kit (6400563-355) no longer available. - Spare part number for o-ring in motors axis 1, 2 and 3 changed. (Is 20406710-439, was 3327520-17)
S	- Published in release R17.2. The following updates are made in this revision: - New tool for replacement of parallel bar. - Updated maintenance procedures. - Added calibration requirements.

B. Contact Information

Contact Information

Department	Contact
Technical Support	support@forgis.com

Department	Contact
Sales	sales@forgis.com
Service	service@forgis.com
Emergency	+1-800-XXX-XXXX